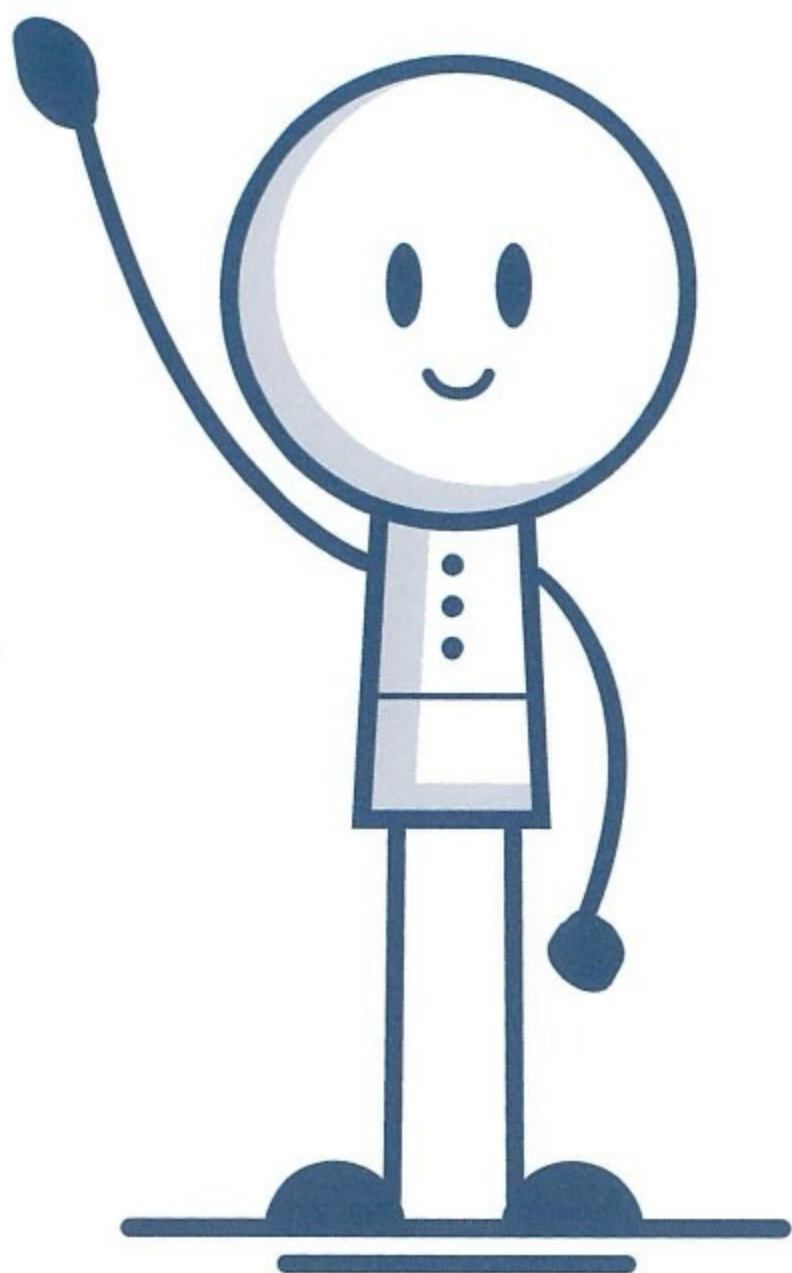




Contents

Why brain-breaks?.....	6
Brain-breaks & engagement.....	7
Brain-breaks & relationships.....	9
Accompanying notes.....	11
To clap or not to clap.....	16
Gotcha.....	17
We hate diagonals.....	18
Look Down, Look Up!.....	19
One, Two, Three Three - Aghh!.....	20
Boi-ing 7.....	21
Find the Fantale.....	22
Seven-11.....	23
What do you look like when?.....	24
Thumb challenge.....	26
Quick clap.....	28
Heads or Tails.....	29
Air performances.....	30
Pop-up facts.....	32
Eraser race.....	34
Line-up (and fold around).....	36
Whole-body Scissors, Paper, Rock.....	38
Back to back.....	40
Flip it!.....	42
Handstacking races.....	43
Keep me up.....	44
Hand clapping games.....	45
What's back on TV?.....	46
Rain circle.....	48
References.....	51

Why brain-breaks?

The experience of school for many students today is dominated by sitting. For much of their day, students are expected to sit still and learn. Whilst it is well-established that sedentary behaviour is associated with a range of health risks such as obesity and cardiac failure, neuroscience has also uncovered many reasons why we might question such a sedentary approach to learning.

Brain-breaks involve taking a short break from in a lesson to engage in a burst of physical movement or a fun mental challenge. Pioneer of brain-based learning, Eric Jensen (2000) explains that movement has a positive effect on learning by:

- increasing circulation, thereby moving more oxygen to areas of the brain for learning
- supporting brain maturation processes such as pruning (eliminating unnecessary synapses), neurogenesis (new cell growth) and myelination (strengthening of neural pathways)
- allowing the brain a break from processing new content
- regulating neurotransmitters such as dopamine and glutamate to enhance focus, mood and learning

Brain-breaks can increase student engagement

Studies over the last 15 years have shown promising results, suggesting brain-breaks have multiple benefits for student learning.

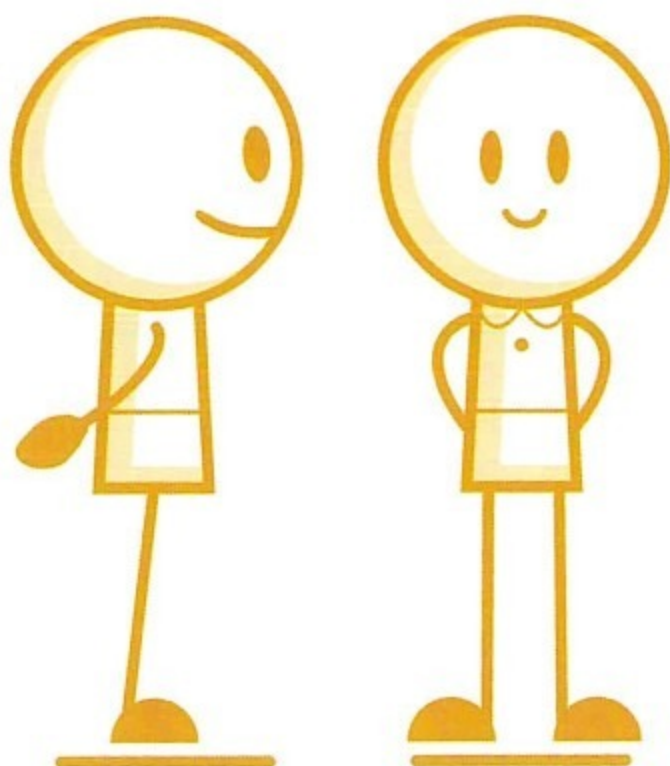
A multitude of studies has shown that regular brain-breaks across the school day are found to enhance students' cognitive functioning and ability to sustain focus for academic work (Donnelly et al., 2009; Grieco, Jowers & Bartholomew, 2009; Ma, Mare & Gurd, 2014; Mahar, et al., 2006; Mahar, 2011).

Importantly, brain-breaks only take up small amount of lesson time yet the benefits of increased engagement levels and cognitive functioning have been found to significantly increase academic performance. Research has shown that students participating in regular brain-breaks display improved academic performance in core subjects like Mathematics and English, as well as academic standardised tests (Álvarez-Bueno, et al., 2016; Ahamed, et al., 2007).

Additionally, brain-breaks have been shown to increase students' perceived competence and raise classroom motivation levels (Vazou et al., 2012).

Brain-breaks decrease students' sedentary time and increase their overall level of physical activity across the day. Studies have also shown brain-breaks can have a positive influence on students' Body Mass Index (BMI) and can support students to maintaining healthy weight (Liu, et al., 2008).

Finally, and not surprisingly, brain-breaks have been shown to improve classroom behaviour and decrease student referrals for discipline issues (Sibley, et al., 2008).



Brain-breaks can strengthen teacher-student relationships

By providing students with a social and fun break in a lesson, there is an increased opportunity and a new context for strengthening teacher-student relationships.

Classrooms are dynamic environments where interactions between students and teachers continuously and powerfully affect learning, wellbeing and development.

Brain-breaks alter the classroom climate by introducing a new collective action. Such activities have been shown to increase students' positive emotions and enjoyment within the classroom (Howie et al., 2014; Vazou, 2014).

Whilst there is no known research investigating the direct link between brain-breaks and teacher-student relationships, there is mounting evidence to suggest that they are positively correlated in a bi-directional fashion. Research has clearly shown that brain-breaks are likely to increase student engagement, and when students are more engaged, teachers become more engaged, helping to nurture positive relationships (Hamre & Pianta, 2005).

The positive student outcomes linked to regular brain-breaks such as enjoyment, positive affect, improved classroom climate and intrinsic motivation, all have the potential to contribute to a caring community of learners.

As students progress through school, relationships with their teachers tend to become more distant and less comfortable, particularly for those students with low intrinsic motivation (Harter, 1996).



Research has suggested that students have more positive perceptions of their teachers when they become more involved in their social environments (Skinner & Belmont, 1993).

Brain-breaks, therefore, offer a possible way to protect students against declines in motivation and feelings of disconnection that some students may experience.

Accompanying notes

Age levels

We have not included recommended age levels as the following brain-breaks can be adapted to be suitable for almost all school-aged children. Younger kids and older kids alike can most definitely reap the benefits of participating in the following brain-breaks...and so can teachers!

Duration

We have deliberately chosen brain-breaks which require minimal, if any, equipment and can be completed within a few minutes. We have not included any activity which requires more than five minutes to introduce and run.

Implementation

The following brain-breaks are designed to be used as a short energiser during a lesson. They can also be used successfully as lesson starters or lesson closers.

Variations

We warmly encourage educators and students to enjoy the following brain-breaks and then to have fun designing innovative variations that can add freshness and ongoing wellbeing and learning benefits.

Tips

Brain-breaks are a low-cost, high-impact intervention for schools. Little training or additional resources are required to start building brain-breaks into lessons.

To maximise the positive impact of brain-breaks, here are some tips and insights that teachers have shared with us:

- Ensure that the activities are developmentally appropriate to students' age and not perceived as 'babyish'
- Help teachers to see brain-breaks as an opportunity to enhance student learning rather than compromising learning time
- Ask students for feedback about their experience of brain-breaks and seek their input and ideas to help fine tune and develop the activities
- Brain-breaks are best implemented within supportive school environments (Webster, et al., 2015)

Please google other brain-break ideas too. There are some wonderful resources out there - many of which are free!

And we'd love for you to get in touch and share any cool new brain-breaks with us:
institute@ggs.vic.edu.au. Thanks!

Role of the teacher

The following suggestions are mentioned to assist you in running brain-breaks as effectively and successfully as possible.

- **Be enthusiastic!**

Your enthusiasm, or lack of it, is contagious and will impact student participation and buy-in.

- **Expect that you will be able to inspire your students to enjoy these fun breaks**

All of these tasks are relatively simple and unsophisticated. That doesn't mean that your students can't love them and reap the many benefits from participating in them.

- **Ensure a sense of belonging**

Remember that one important aim for incorporating brain-breaks is to help students develop a sense of belonging to the group.

- **Keep the rules to a minimum.**

Explain the rules as briefly and as clearly as possible.



Disclaimer

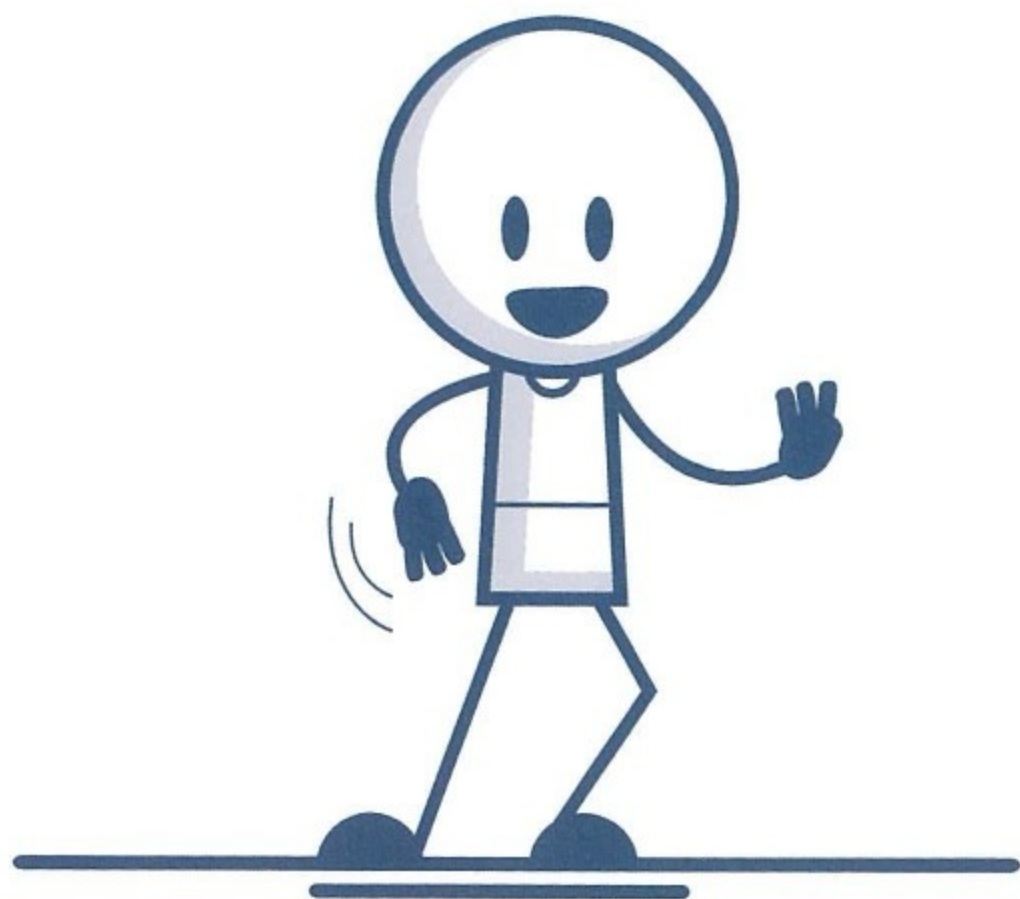
Each of the brain-breaks described in this book have been successfully applied by teachers at Geelong Grammar School (GGS).

Trainers at GGS's Institute of Positive Education have also harnessed the power of these activities with educators and students around the world, and there certainly appears to be a universal appeal to participating in fun, inclusive and stimulating brain-breaks.

Despite our successes in using the activities described in this booklet, we encourage educators to consider, with prudence, when, where, why and how to best use each activity.

Not all activities will work for you and your students. Also, be mindful to harness your intuition and experience to skillfully introduce a new activity and draw upon your classroom management skills to ensure the desired outcomes are achieved.

Many of these activities have been handed down through generations of educators so have fun with them! We hope that they stimulate your students' engagement levels and strengthen your relationship with your students.



1. To clap or not to clap

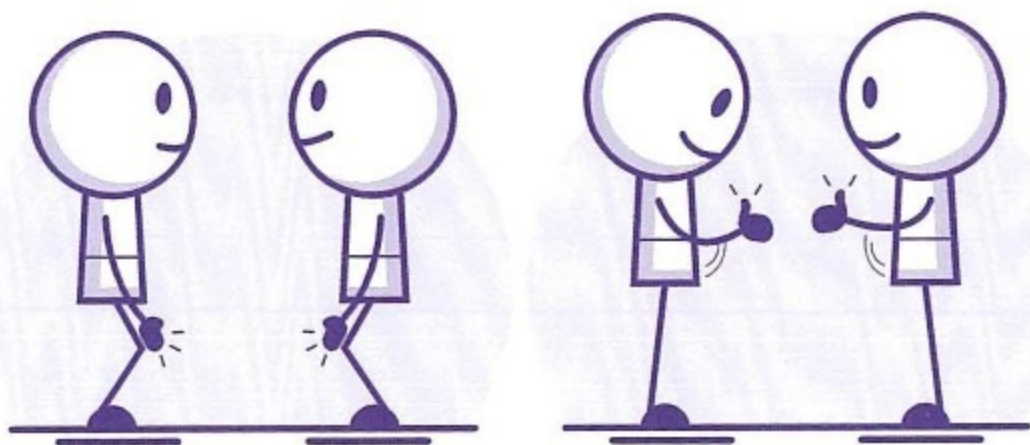
Energy: Medium
Equipment: None
Duration: 2 minutes

Students stand in pairs facing one another.

Simultaneously, Student A and B slap their own thighs and then independently make their own decision as they raise their hands to either show thumbs pointing left, up or right.

If Student A and B make the same decision, they again slap their thighs and follow with a clap!

If Student A and B make different decisions, they again slap their thighs, but they don't get to clap! Instead they simply make another decision as they raise their hands once more.



2. Gotcha!

Energy: Medium
Equipment: None
Duration: 3 minutes

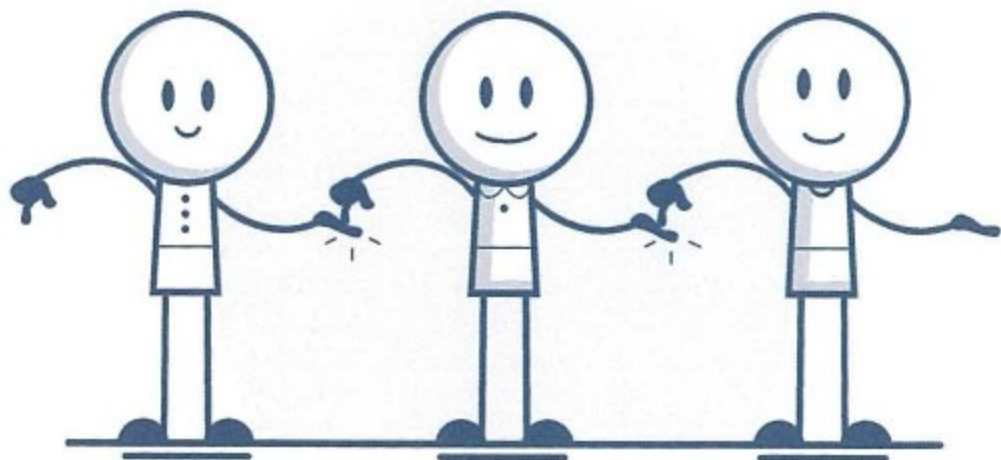
Students stand in a circle facing inwards.

Students place their left hand out flat, palm up, to their left side.

Students point their right index finger and place it into the outstretched palm of the person to their right.

On “Go!”, students attempt to grasp the index finger of the student on their left, whilst simultaneously escape the attempted grasp of the student on their right.

“Go!”



3. We hate diagonals

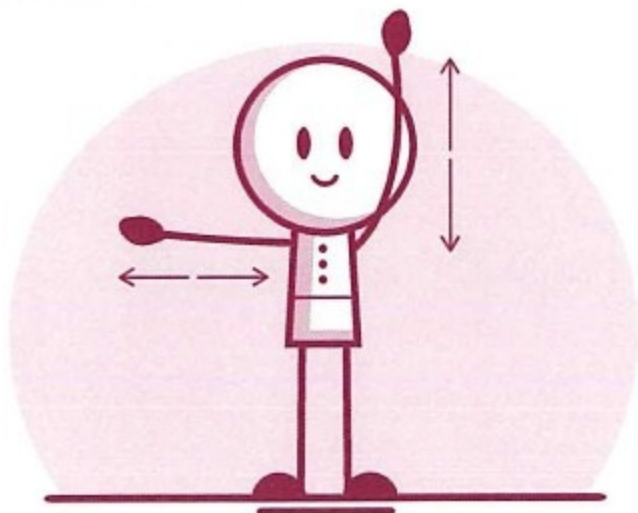
Energy: Medium
Equipment: None
Duration: 2 minutes

Students raise their left hand in the air, ensuring they move it only up and down the 'vertical axis'.

Students then take their right hand, ensuring they move it only side to side, along the 'horizontal axis'.

Students then combine these two movements, whilst at the same time ensuring no diagonal movements creep in. Remember, 'we hate diagonals'!

Once mastered, swap sides - right hand vertical axis, left hand horizontal axis.



4. Look Down, Look Up!

Energy: Low
Equipment: Small object
Duration: 5 minutes

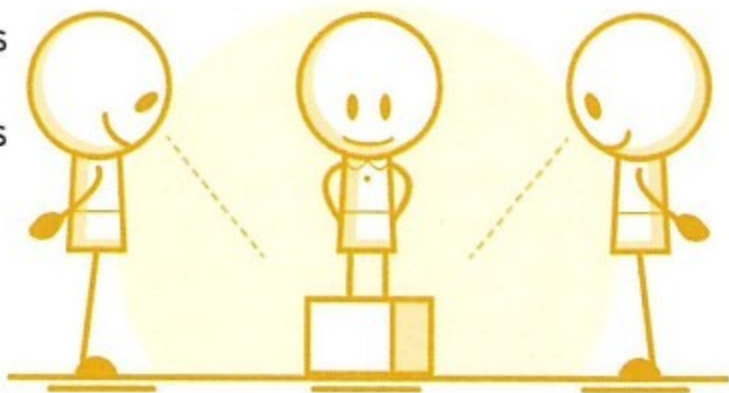
Students stand in a circle facing inwards. Groups of approximately 10 – 12 are ideal.

Students stare at a small object placed on the floor in the middle of the circle.

Each student decides on a person that they will directly look at when instructed to 'Look Up'. If they find that this person has also chosen them and is looking directly back at them, then both students step out of the circle.

After each turn, students are asked to 'Look Down'. They return their gaze to the floor and decide who they will look at as they await the next call of 'Look Up'.

Play continues until only one or two players remain.

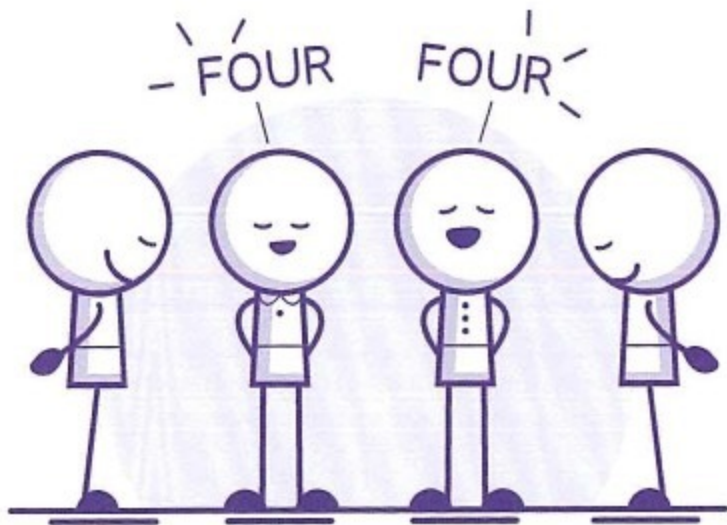


5. One, Two, Three Three - Aghh!

Energy: Low
Equipment: None
Duration: 3 minutes

Students stand and close their eyes. Any student can begin this brain-break by calling out “one”. Another student must then call out “two” and so on in numerical order until all students have called out a number.

If two students call out the same number at the same time, the count resets and the game begins again.



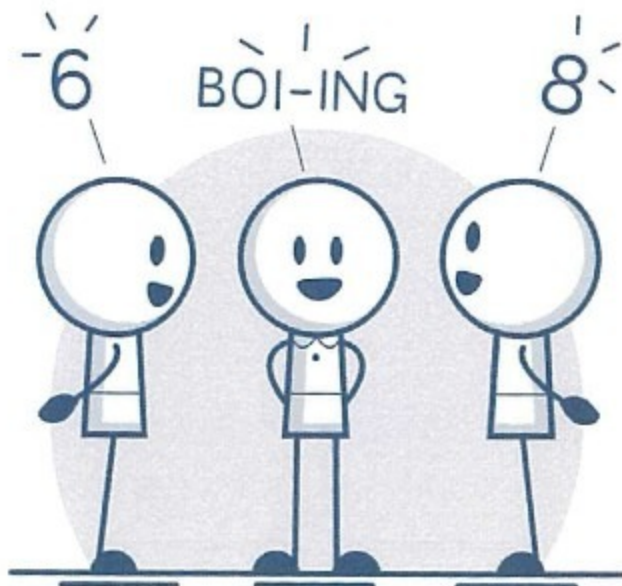
Variation: Students take turns in standing up, with only three rules to follow. Someone must always be standing, no more than five people can be standing at one time and no one can stay standing for more than five seconds.

6. Boi-ing 7

Energy: Low
Equipment: None
Duration: 4 minutes

Students stand in a circle facing inwards. A nominated student begins by saying “one”. The student to the left of the first one says “two” and so on around the circle, calling out numbers in numerical order.

Every time a number comes along that contains a “7” or is a multiple of 7, students must call out “boi-ing” (a jumping sound because “7” is being jumped!) in place of the actual number. Counting then continues as normal until the next “7”...
...be careful!!



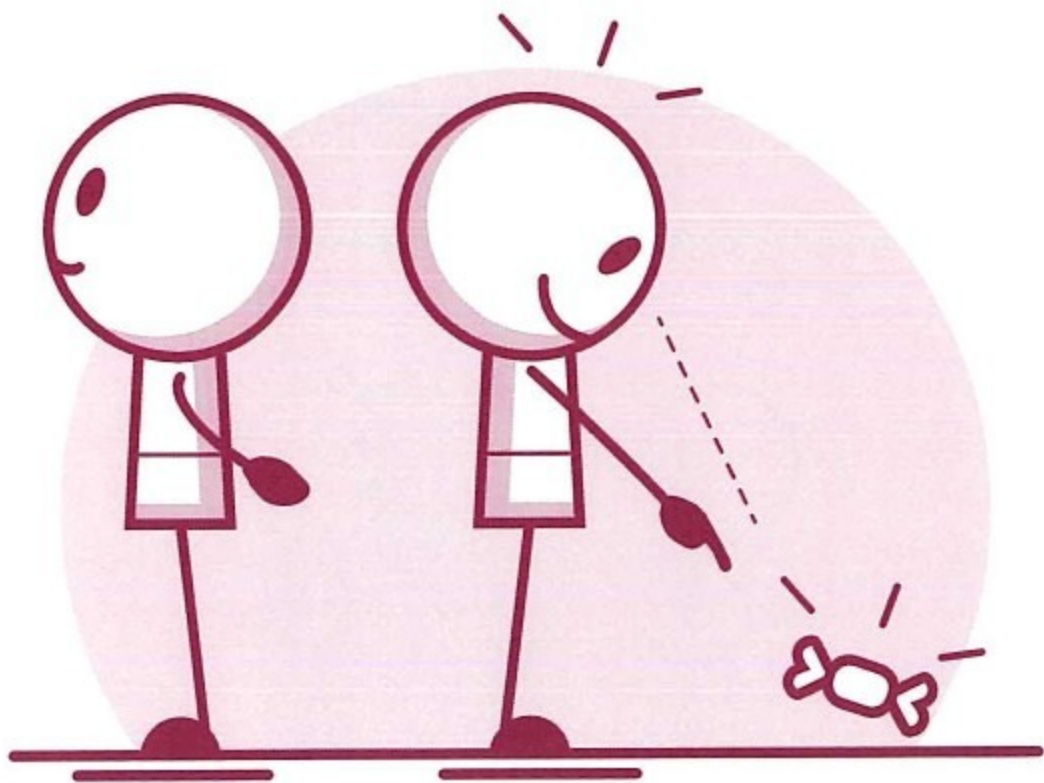
7. Find the Fantale

Energy: Medium
Equipment: 1 Fantale
Duration: 1 minute

Prior to the lesson beginning, the teacher carefully hides a Fantale somewhere in the room.

At a random point in the lesson, the teacher calls out "Find the Fantale!". Students hunt for the Fantale (in a controlled fashion!).

The end.



8. Seven-11

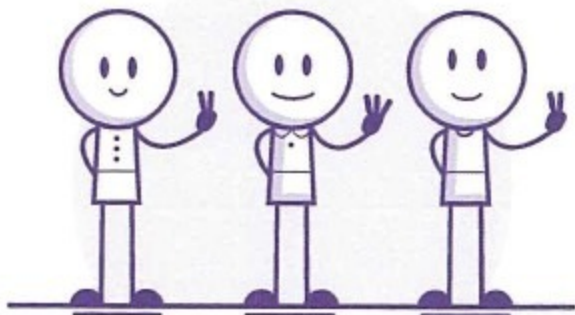
Energy: Low
Equipment: None
Duration: 2 minutes

Students stand facing one another in groups of three. One of the players says, "One, two, three, Show!"

On 'Show', all three students throw out as many fingers (0 – 5) on one hand as they deem necessary to come up with the solution, which in this case is to achieve the number seven.

In other words, the hope is that, *without verbal communication*, the three students can display a total number of seven fingers. How many times will it take your students to show 7?

$$2 + 3 + 2 = 7$$



A two handed variation can be used where the goal to be reached is eleven!

9. What do you look like when?

Energy: Medium
Equipment: None
Duration: 3 minutes

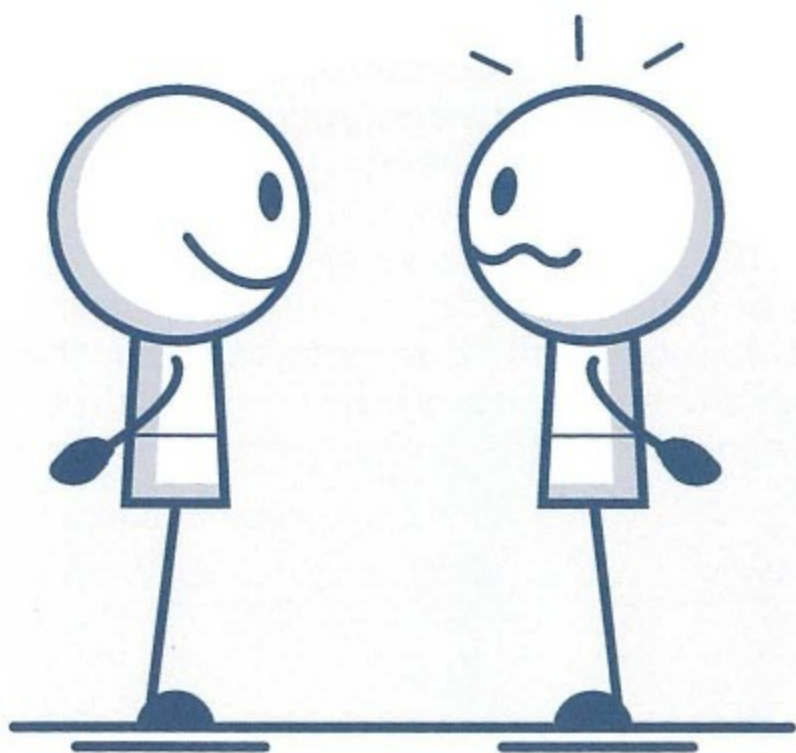
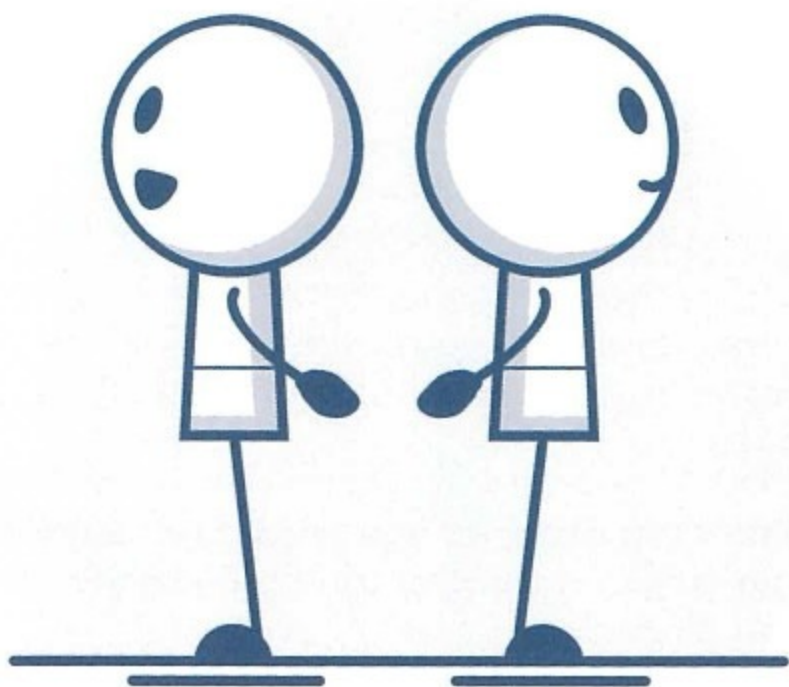
Students stand back to back in pairs. Either partner initiates the action using the phrase: *"What you look like when...."*

After two seconds of silent moment of facial preparation, each player spins 180° and faces one another with their best facial expression depicting the moment described.

Partners may like to have several goes.

'What do you look like when...' examples include:

1. ... you wake up in the morning?
2. ... you get an A on a Maths test?
3. ... you find out your teacher is away?
4. ... your pet wees on the carpet?
5. ... the pilot says to expect turbulence?



10. Thumb challenge

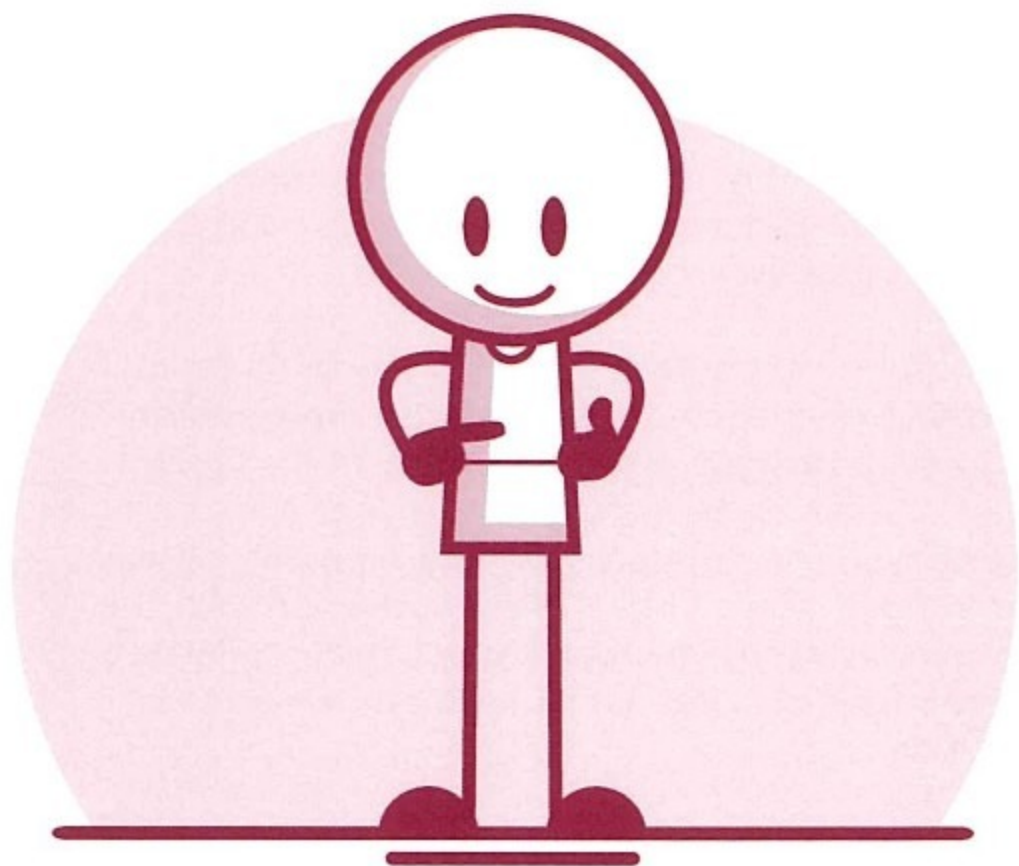
Energy: Low
Equipment: None
Duration: 1 minute

Students stand and watch the teacher demonstrate the dexterous thumb pointing challenge.

Students then attempt the challenge, having multiple turns and trying to improve their speed and coordination.

To perform the thumb pointing challenge – with one hand, simply clench your fist and give a ‘thumbs up’ sign. With your other hand, keep your thumb tucked in and point your pointer finger directly at the raised thumb.

On ‘switch’, simply reverse positions, with the opposite hand now pointing at the opposite raised thumb. Simply alternate between these two positions and you are doing the thumb challenge!



11. Quick clap

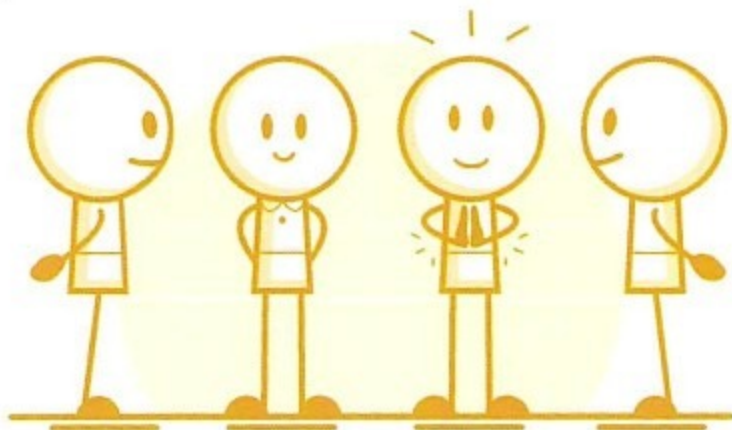
Energy: Med
Equipment: None
Duration: 4 minute

Students stand in a circle facing inwards.

The teacher, or a student, starts by clapping their hands together. The person immediately to the right then quickly claps their hands, and on the clap progresses around the circle.

It can be very effective for the clap to be a double or a triple clap, with the next person starting straight after the person's first clap.

It can be fun to start multiple clapping chains and have them chase one another around the circle. Alternatively send claps off in different directions and see what happens when they collide.



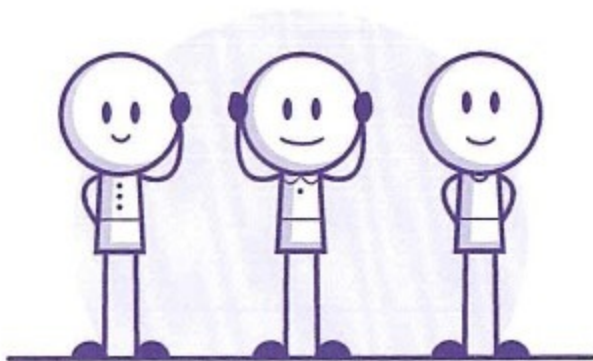
12. Heads or Tails

Energy: Low
Equipment: Two coins
Duration: 3 minutes

Students stand and assume one of the following three poses:

- Heads: Both hands on their head
- Tails: Both hands on their backside
- Combo: One hand on head, one hand on backside

Two coins are then tossed into the air, two-up style. Students in the correct pose, as shown by the two landed coins, are able to remain standing. Students in the incorrect pose take a seat.



Remaining students choose their next pose, which can be the same or different to their last pose, and hope that the coins land once again in their favour!

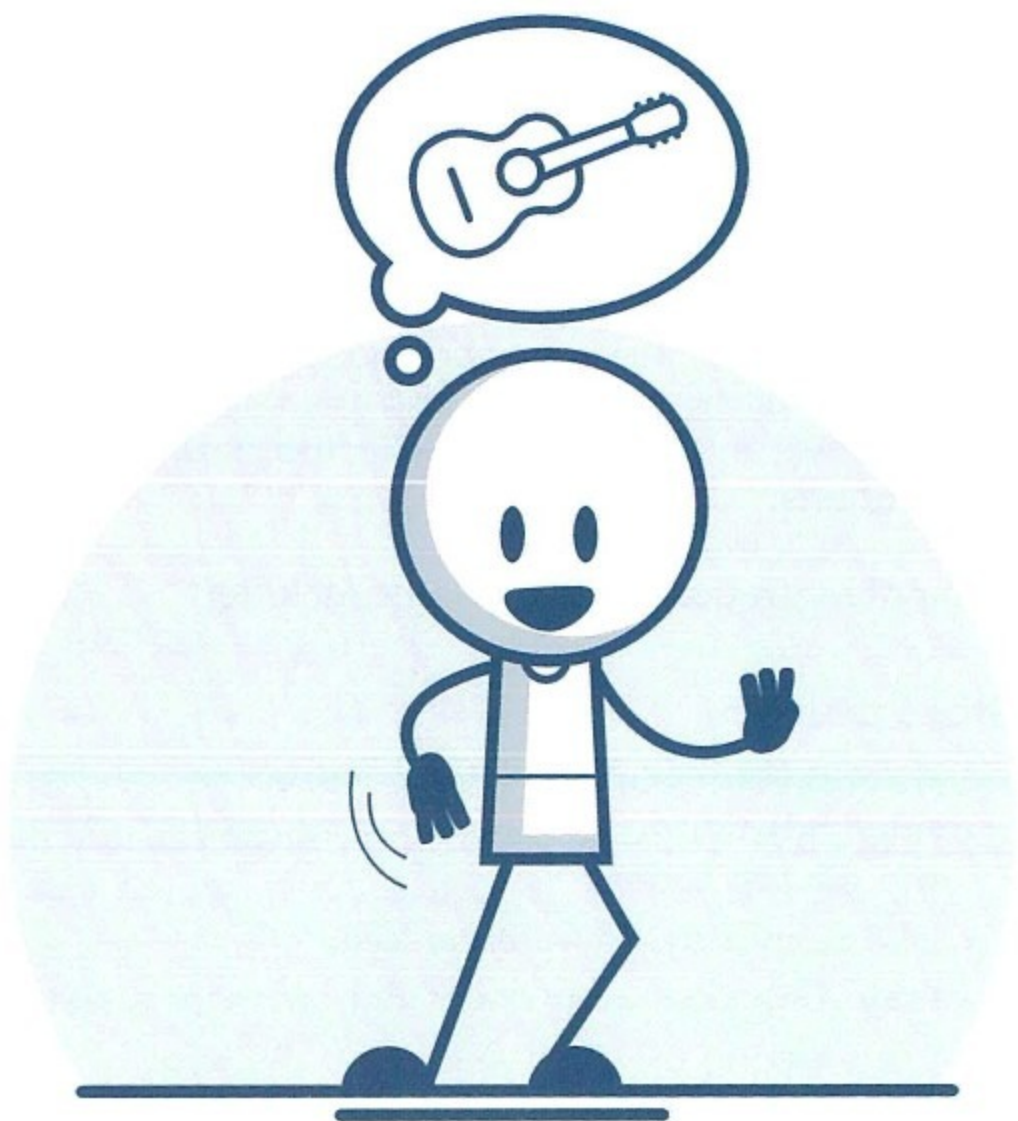
13. Air performances

Energy: High
Equipment: None
Duration: 2 minutes

Students stand and await the instructions to which air activity (aka virtual or imaginary activity) they will be instructed to complete. Possibilities include:

- Air guitar (with no guitar)
- Air skipping (with no rope)
- Air juggling (with no balls)
- Air tennis (with no racquets)
- Air piano playing (with no piano)
- Air hackey sack (with no sack)

Students give their very best virtual performances and enjoy the feeling of outstanding imaginary success!



14. Pop-up facts

Energy: Medium

Equipment: None

Duration: 3 minutes

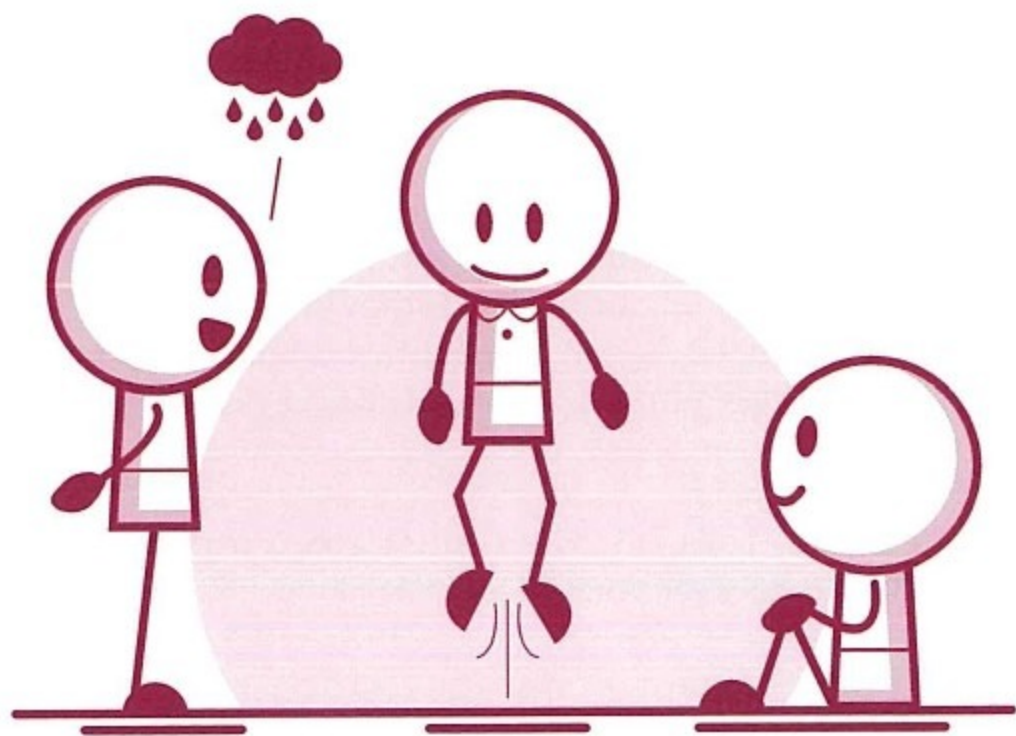
Students sit at their desks or on the floor in a circle. Students take it in turns calling out a category and an action.

All those who fit the category pop-up in their place, complete the action and then sit back down. Have and be creative with the categories and actions.

Example categories and actions include:

Stand up if you:

- Have a brother & click your fingers
- Have lived in more than three houses & punch the air five times
- Like rainy days & wink an eye
- Play a musical instrument & beat your chest
- Are wearing a watch & hop on the spot
- Made your bed this morning & take a bow



15. Eraser race

Energy: Medium

Equipment: Two whiteboard erasers

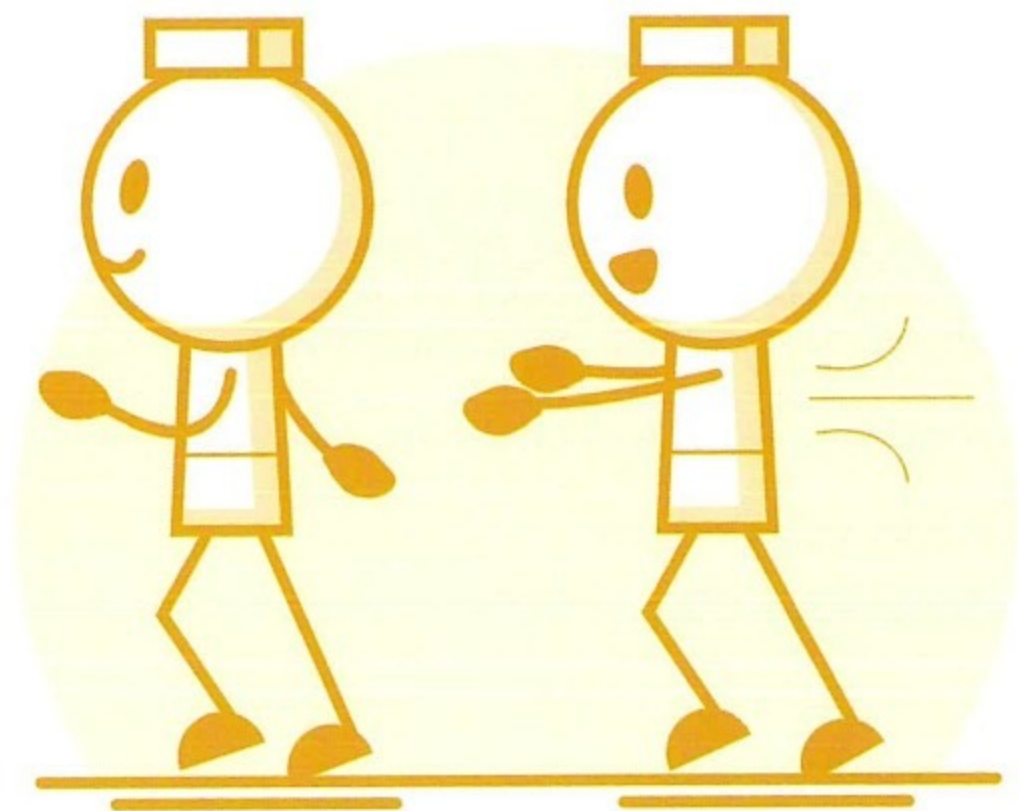
Duration: 3 minutes

Students sit at their desk or on the floor. Two students are nominated and handed a whiteboard eraser (or similar soft object). Both students place the whiteboard eraser on their heads.

Decide which student is the Eraser Racer and which one is the Eraser Chaser. The Eraser Racer is given several seconds to get away and then the *walking* race is on! Students cannot touch the eraser on their head as they move around the classroom.

If the eraser falls off, they must stop, replace the eraser before starting off again.

The Eraser Racer is endeavouring to walk one lap and return to their empty place without being caught by the Eraser Chaser. New students have a go for the next round.



16. Line-up (and fold around)

Energy: Medium
Equipment: None
Duration: 3 minutes

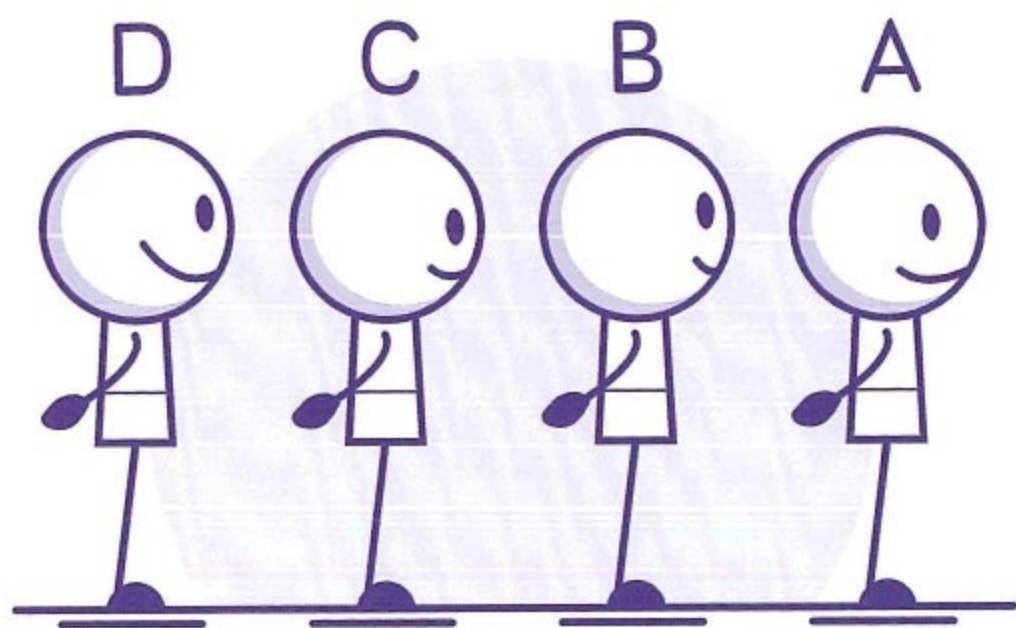
Provide students with an instruction to arrange themselves in one orderly line according to the theme provided.

At the end of the line-up, the head of the line could fold around to the tail and suddenly all class members are in random pairs - ready perhaps for a follow-on lesson activity.

Example line-up instructions include:

Line-up according to:

- Birthday (Day of the year)
- Alphabetical order of middle names
- Time you got out of bed this morning
- The length of your hair
- The number of people in your family
- Your locker number or phone number



17. Whole-body Scissors, Paper, Rock

Energy: Medium
Equipment: None
Duration: 3 minutes

Everyone knows how to play the classic hand game – Scissors, Paper, Rock. This variation is a little bigger...you have to use your whole-body to perform the action!

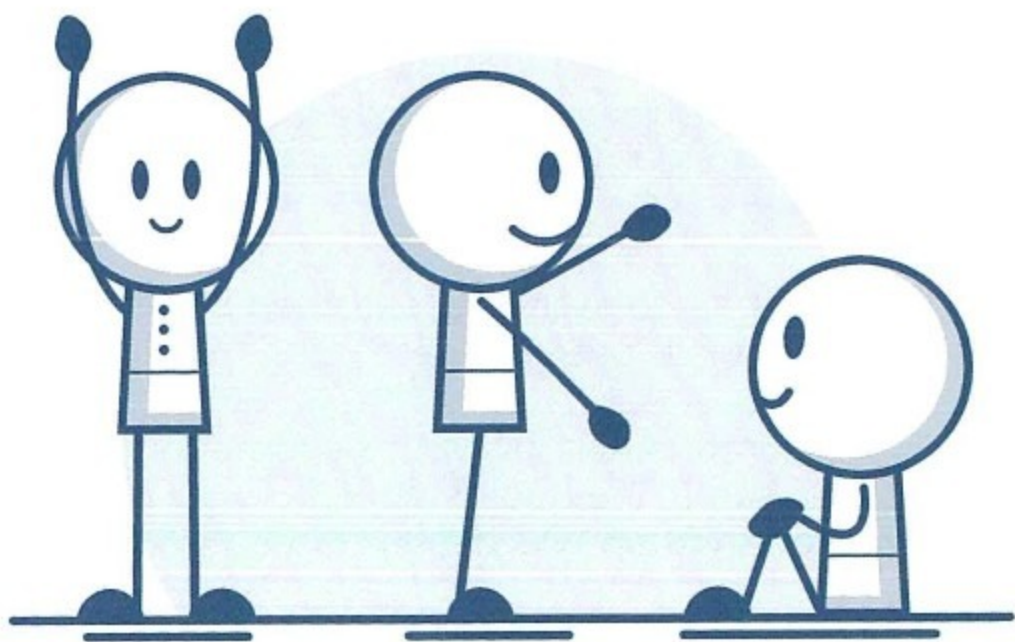
In pairs, students prepare for each round by counting in with the words: “scissors, paper, rock” before making their chosen shape.

For scissors, your two moving arms become the scissors.

For paper, you stretch your body out as straight as possible, with arms raised above your head.

For rock, you scrunch yourself up into a ball.

Have fun, throwing your whole body into this old-time favourite.



18. Back to back

Energy: Medium

Equipment: None

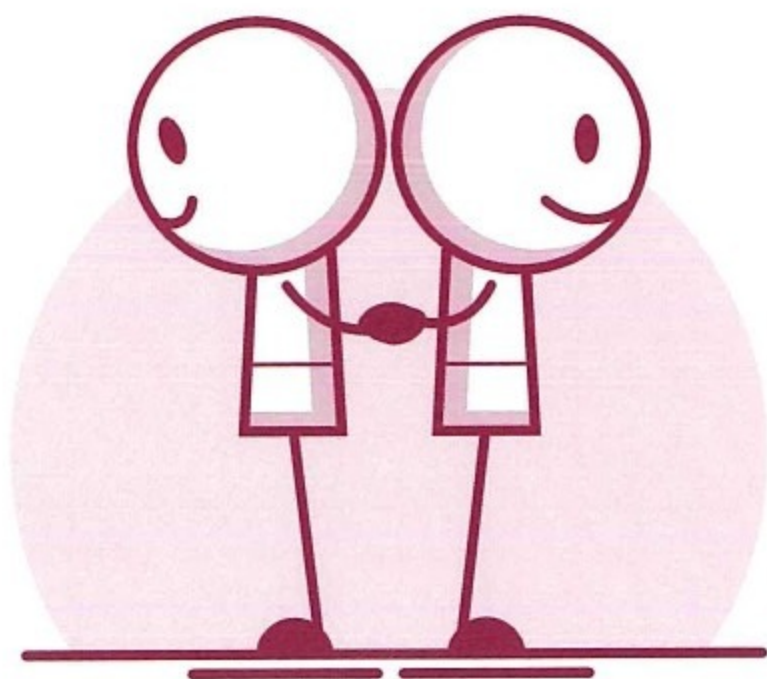
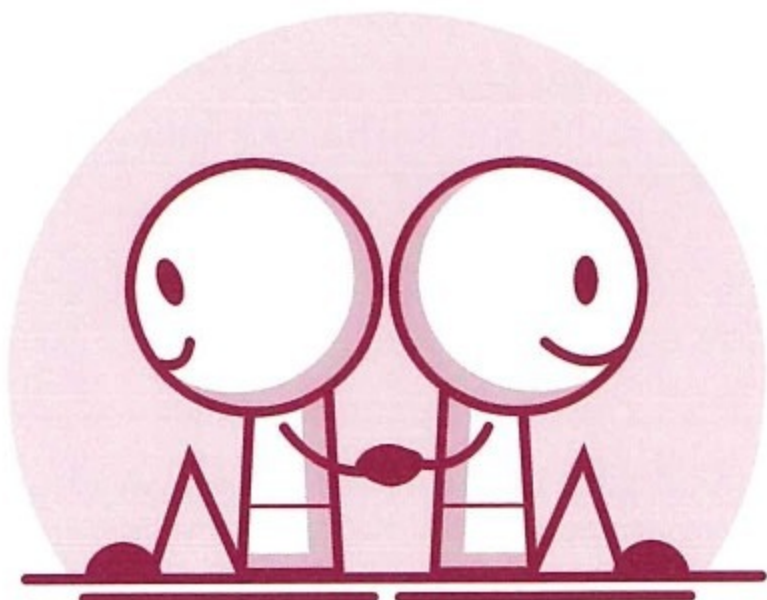
Duration: 2 minutes

Students arrange themselves in similar height pairs. Each pair sits down back to back.

Once the pairs are sitting back to back, they link arms and then attempt the challenge of working together to stand up without using their arms or hands to help.

Challenge a pair to stand up and then sit back down five times in one minute.

Challenge some expert pairs to become larger groups and see if they can still find a way for them all to stand up whilst their arms remain linked.



19. Flip it!

Energy: Medium

Equipment: Plastic bottle and other objects

Duration: 3 minutes

Embrace the craze of Bottle Flipping and challenge students to successfully flip a water bottle.

Allow the students to design their own challenges by placing limits on the throw (non-preferred hand, eyes closed, etc) and by suggesting different landing zones (floor, desk, window sill, door handle!)



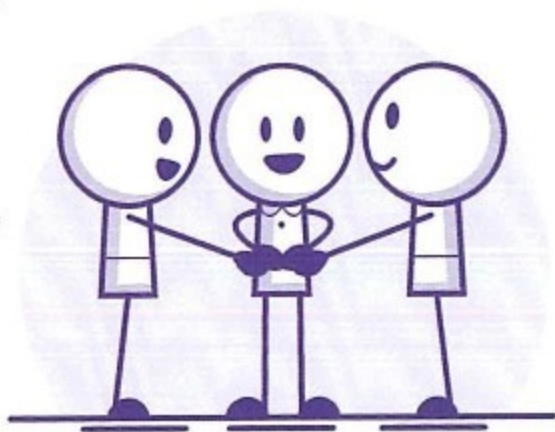
20. Handstacking races

Energy:	Medium
Equipment:	None
Duration:	2 minutes

To begin the handstacking race, divide students into two teams, or alternatively divide the class into teams of six.

Each team forms a circle and stack their right hands on top of each other in the middle of the circle. Next, have them stack their left hands on top of the right hands.

On 'go', the player whose hand is on the bottom moves his or her hand to the top of the stack. The other players follow, moving one hand at a time.



Once teams have practiced maneuvering their hands, let the race begin! Try one lap races, two lap races or more. Mark the top player's hand in some way to help signal the end of a lap!

21. Keep me up

Energy: Medium
Equipment: 1 or 2 balloons
Duration: 3 minutes

Challenge the students to simply keep a blown-up balloon in the air for as long as possible. They can use any body part, as long as the balloon does not touch the ground/carpet (and/or walls/ceiling).

By all means challenge the class with multiple balloons.

What would this look like if you had one balloon that the students could keep up in the air for an extended period of time, while the lesson continued?



22. Hand clapping games

Energy: Medium
Equipment: None
Duration: 3 minutes

Students work in pairs, or small groups or one large class group.

Students are shown a particular clapping rhythm, which they then emulate.

Alternatively, students design their own specific clapping routine which they rehearse and perform.

You may wish to search on the internet and/or YouTube for some excellent hand clapping resources and videos.



23. What's back on TV?

Energy: Medium
Equipment: None
Duration: 3 minutes

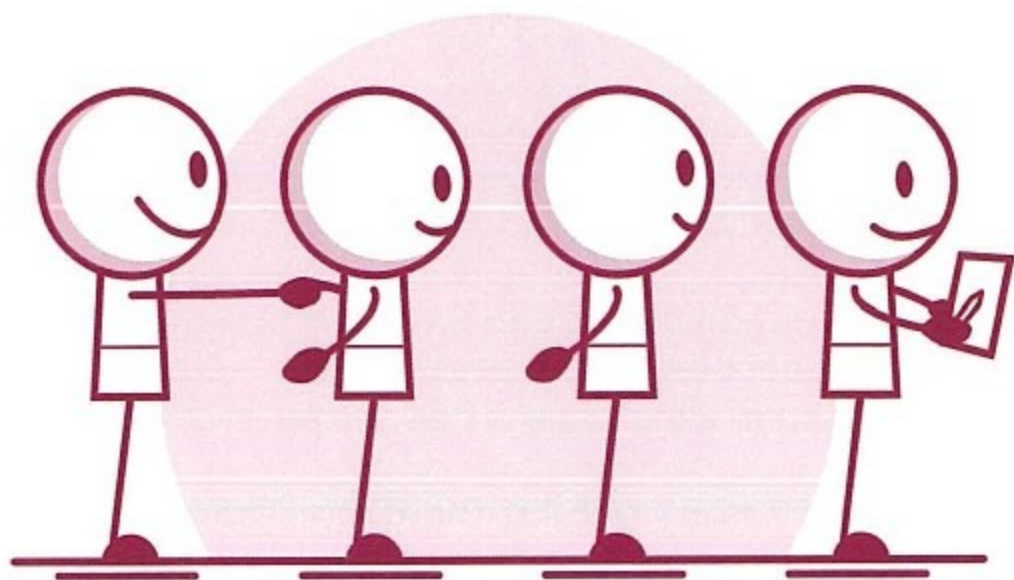
Divide students into teams of six to eight.

Each team sits in a straight row, one team member behind the other. The first person in the line is given pen and paper.

Each team is like a transmission station. Team members try to see how quickly and accurately they can pass along a picture by drawing it with a forefinger on each others' backs.

Decide on a simple picture for the teams to draw and show this image to the back person in each team. Teams then send the transmission, with it concluding with the front person drawing the image they felt drawn on their back on to the supplied piece of paper.

Compare times and accuracy (and have a laugh).



24. Rain circle

Energy: High
Equipment: None
Duration: 4 minutes

Students stand in a circle facing inwards.

The teacher, or a student, starts by rubbing their hands together, to signal the start of the rain.

The person immediately to the right then starts rubbing his/her hands together and this action progresses quickly around the circle.

Once the action returns to where it started, the leader introduces the second action – clicking fingers, to represent the pitter patter of the rain.

This action moves around the circle, followed by thigh slapping and then foot stamping to signify the increasing magnitude of the rainstorm.

Reverse the actions and bring the sound back to a gentle swishing of the rubbing hands.





References

Ahamed, Y., Macdonald, H., Reed, K., Naylor, P. J., Liu-Ambrose, T., & McKay, H. (2007). School-based physical activity does not compromise children's academic performance. *Medicine and science in sports and exercise*, 39(2), 371.

Álvarez-Bueno, C., Pesce, C., Cervero-Redondo, I., Sánchez-López, M., Pardo-Guijarro, M. J., & Martínez-Vizcaíno, V. (2016). Association of physical activity with cognition, metacognition and academic performance in children and adolescents: a protocol for systematic review and meta-analysis. *BMJ open*, 6(6), e011065.

Donnelly, J. E., Greene, J. L., Gibson, C. A., Smith, B. K., Washburn, R. A., Sullivan, D. K., ... & Jacobsen, D. J. (2009). Physical Activity Across the Curriculum (PAAC): a randomized controlled trial to promote physical activity and diminish overweight and obesity in elementary school children. *Preventive medicine*, 49(4), 336-341.

Grieco, L. A., Jowers, E. M., & Bartholomew, J. B. (2009). Physically active academic lessons and time on task: the moderating effect of body mass index. *Med Sci Sports Exerc*, 41(10), 1921-1926.

Hamre, B. K., & Pianta, R. C. (2005). Can instructional and emotional support in the first grade classroom make a difference for children at risk of school failure?. *Child development*, 76(5), 949-967.

Harter, S. (1996). Teacher and classmate influences on scholastic motivation, self-esteem, and level of voice in adolescents. In J. Juvonen & K. Wentzel (Eds.), *Social motivation: Understanding children's school adjustment*. New York: Cambridge University Press.

Howie, E. K., Newman-Norlund, R. D., & Pate, R. R. (2014). Smiles count but minutes matter: responses to classroom exercise breaks. *American journal of health behavior*, 38(5), 681-689.

Jensen, E. (2000). Moving with the brain in mind. *Educational leadership*, 58(3), 34-38.

Liu, A., Hu, X., Ma, G., Cui, Z., Pan, Y., Chang, S., & Chen, C. (2008). Evaluation of a classroom based physical activity promoting programme. *Obesity Reviews*, 9(s1), 130-134.

Ma, J. K., Mare, L. L., & Gurd, B. J. (2014). Classroom-based high-intensity interval activity improves off-task behaviour in primary school students. *Applied Physiology, Nutrition, and Metabolism*, 39(12), 1332-1337.

Mahar, M. T. (2011). Impact of short bouts of physical activity on attention-to-task in elementary school children. *Preventive medicine*, 52, S60-S64.

This booklet

This booklet is designed to be a helpful resource for you. It's not a specific programme to follow but rather, it's a starting point for thinking about how to integrate physical movement into your lessons. We hope you enjoy taking brain-breaks with your students and sharing your learning and ideas with other teachers.

